

CN PA2----- UDPPING----

In this assignment we had to change the packet being sent by application layer from UDP to ICMP for sending an ICMP ping. We stored a magic no DECAF in payload to identify ICMP reply. We also stored source port for receiving UDP reply later in ICMP "id" field. After iCMP reply, we converted it back into UDP.

Changes-

In the e1000_xmit_frame we checked the daddr of iph header and if it is 100.100.100.100 we changed the packet to ICMP. We compute the checksums of IP header and ICMP header in compute_chksum, compute_chksum2. We sent the packet. We have printed appropriate kernel debug messages.

While receiving the ICMP echo reply, we have changed the e1000_receive_skb function. We checked if the packet protocol is ICMP. Then if it has a magic number. Then we changed the ip hdr and packet protocol manually. We changed the iphdr packet protocol back to 17(UDP). We compute the checksum of IP header manually as in above part. We then compute the checksum of UDP header.

For it we have written a pseudo_ip_header to append it to the start of UDP header. We stored UDP length in pseudo_header->tot_len. Then we calculated checksum and put it in udphdr->check. We sent a UDP ping by the application ./udpping and got an output of Congrats test Passed indicating that our UDP ping was sent and received successfully.

b.) Yes, I am getting the expected output after pinging Google using udpping.

c.) Yes, the ping to Google using the ping application works correctly after a Successful ping to Google using udping.

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Congrats: test passed
ww.google.comork@vidush-does-work-Standard-PC-i440
PING www.google.com (216.58.200.164) 56(84) bytes
64 bytes from del11s06-in-f4.1e100.net (216.58.200.164): icmp_seq=1 ttl=64 time=3.516 ms
64 bytes from del11s06-in-f4.1e100.net (216.58.200.164): icmp_seq=2 ttl=64 time=3.887 ms
^C
--- www.google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss,
rtt min/avg/max/mdev = 3.516/3.887/4.259/0.371 ms
ing 216.58.200.164idush-does-work-Standard-PC-i440
UDP echo: 216.58.200.164
Congrats: test passed
vidush-does-work@vidush-does-work-Standard-PC-i440
```

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